Aliah University

Paper Code: CSEUGPC12, Paper name: Design and Analysis of Algorithms

BTECH

Full Marks: 80; Time: 3 Hrs.

(The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as possible)

Group-A

Answer any 5 questions $5 \times 1 = 5$

1. Dijkstra's Algorithm is used to solve ____________ problems.
   a) All pair shortest path b) Single source shortest path
   c) Network flow d) Sorting

2. Kruskal's algorithm is a ______
   a) divide and conquer algorithm b) dynamic programming algorithm
   c) greedy algorithm d) approximation algorithm

3. What is the average case time complexity of merge sort?
   a) O(n log n) b) O(n2) c) O(n2 log n) d) O(n log n2)

4. What is the median of three techniques in quick sort?
   a) quick sort with random partitions
   b) quick sort with random choice of pivot
   c) choosing median element as pivot
   d) choosing median of first, last and middle element as pivot

5. Which of the following methods can be used to solve the Knapsack problem?
   a) Brute force algorithm b) Recursion
   c) Dynamic programming d) Brute force, Recursion and Dynamic Programming

6. Consider the two matrices P and Q which are 10 x 20 and 20 x 30 matrices respectively. What is the number of multiplications required to multiply the two matrices?
   a) 10*20 b) 20*30 c) 10*30 d) 10*20*30

Group- B

Answer any five (05) questions 5 x 6 =30

1. Define big-O, big-theta notation. Write the graphical interpretation of it.
2. Suppose you have two sorted array of size n. How you get a single sorted array from this two using merging technique?
3. Describe how to solve Knapsack problem using greedy algorithm
4. Describe Breadth First Search and Depth First search of a graph with example.
5. Define spanning tree of graph. Describe Kruskal algorithm for obtaining minimum spanning tree from a graph.
6. What is independent set of a graph? How to find an independent set of tree using greedy technique?

10 40 30 50 M=10

5 4 6 3

Group- C

Answer any three (03) questions 3 x 15 =45

1. Describe matrix chain multiplication problem. Write and explain the recurrence relation for solving matrix chain multiplication problem using dynamic programming technique. solve the following matrix chain multiplication problem using dynamic programming:

   A1:20x5, A2: 5x50, A3:50x10, A4: 10x30 3+ 4+ 8 =15

2. In which complexity class the following function belong? $f(n) = 5n^3 + 6n$. Prove that for any polynomial p(n) of degree k, p(n) = O(n^k). Solve the following recurrence relation by using master theorem:

   T(n)=7T(n/2) + 18n^2. 4 + 6 + 5= 15

3. State the 0/1 Knapsack problem. Consider instance of the 0/1 knapsack problem as below with P depicting the value and w depicting the weight of each item whereas M denotes the total weight carrying capacity of the knapsack. Find optimal answer using dynamic programming technique.

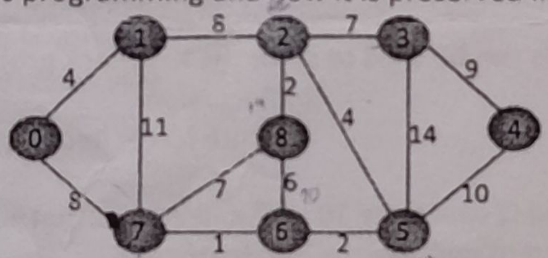
   P: [10 40 30 50] W = [5 4 6 3], M : 10

   If fractional part of items are allowed to take then what will be the optimal solution. 3+ 8 + 4=15

4. Analyse time complexity of quick sort algorithm (write the recurrence relation and solve it). When it achieves quadratic complexity. What is the difference between divide-and-conquer and greedy techniques? Write the generalized equation of the divide and conquer technique.

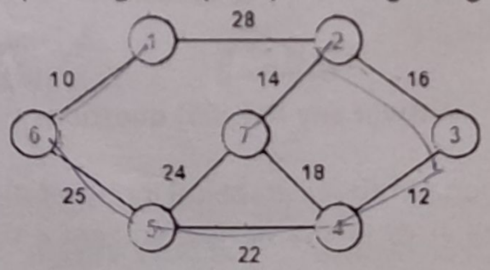
   6 + 2 + 3 + 3=15

5. Describe single source shortest path problem. For the following graph find the shortest paths from the source(vertex 0) to all vertices using Dijkstra algorithm. What is optimal substructure property of dynamic programming and how it is preserved in this algorithm?



   3 + 8 + 4 =15

6. Construct the minimum spanning tree (MST) for the given graph using Prim's Algorithm.



   Is it a greedy technique? If yes then why? Describe the greedy move of the Prim's algorithm. Describe the difference between dynamic programming and divide and conquer technique.

greed → prims → locally optimal solution.

8 + 4 + 3 =15